SERVER CONFIGURATION FOR SUPERMARKET NETWORK

Group 7

Names :	Haikal Raihan Hafidz
	Muhammad Ramadhan Ikhlaashul Abda
	Flora Aulia Nilamaya
Faculty :	Muhammad Riza Iqbal Latief, S.T.
Class :	2CS2

CEP CCIT

FACULTY OF ENGINEERING UNIVERSITY OF INDONESIA 2025

PROJECT INFORMATION

Project Title : Server Configuration for Supermarket Network

Batch Code : 2CS2

Start Date : March 17, 2025

End Date : March 24, 2025

Name of Faculty : Muhammad Riza Iqbal Latief, S.T., M.T.

Names of Developer :

1. Haikal Raihan Hafidz
2. Muhammad Ramadhan Ikhlaashul Abda
3. Flora Aulia Nilamaya

Date of Submission : March 25, 2025

UNIVERSITAS INDONESIA
CEP-CCIT
FAKULTAS TEKNIK

CERTIFICATE

This document serves as verification for the report "Server Configuration for Supermarket Network Structure" written by Haikal Raihan Hafidz, Muhammad Ramadhan Ikhlasul Abda, and Flora Aulia Nilamaya. The goal of this project is to complete the CCIT-FTUI course requirement.

ACKNOWLEDGEMENT

First of all, we want to express our gratitude to God Almighty for its endless giving to us, which has enabled us to attend a university at CCIT-FTUI and be able to carry out and complete this paper without any interruptions. We also want to express our appreciation to our lecturer, Mr. Riza Iqbal Latief, S.T. as a consultant who helped us to finish this paper by giving suggestions and recommendations, without the lecturer and members of group 7, this project would not have been possible.

This paper aims to analyze and implement server configurations—specifically DHCP, DNS, database, and web servers—for a supermarket network. It provides understanding of how these servers contribute to efficient network management, data handling, and service delivery within a supermarket environment.

We realize that the writing of this paper is far from perfect, so that's why we apologize if there is any mistake in our writing, we are very open to receiving suggestions and criticism of this paper. We hope that, aside from the accuracy of the writing, it will not be a problem for a reader to receive the knowledge from our paper.

SYSTEM ANALYSIS

This paper discusses server configuration for supermarket networks with the aim of understanding the limits of our ability as writers to set up and manage servers. This research covers server configurations such as DHCP for dynamic IP allocation, DNS for domain resolution, web servers for online services, and database servers for data management.

In this paper, we also explain the complete steps from the process of downloading the server software, installation, to the testing stage to ensure all configurations work optimally. This research aims to delve deeper into the technical capabilities needed to build a reliable server infrastructure.

INTRODUCTION

In this practice, we implement several types of servers to support network operations in a supermarket environment. Here are the roles of each device in the network:

PC	IP Address	Function	Operating System
PC 1	192.168.10.3/24	DHCP Server	RHEL
PC 2	192.168.10.4/24	DNS Server	RHEL
PC 3	192.168.10.5/24	Web Server	RHEL
PC 4	192.168.10.6/24	DB Server	RHEL
PC 5	DHCP Client	Client Device	Windows XP

1. **PC 1 - DHCP Server (192.168.10.3/24 - RHEL)**
   DHCP Server automatically assigns IP addresses to other devices in the network.

2. **PC 2 - DNS Server (192.168.10.4/24 - RHEL)**
   DNS Server translates domain names into IP addresses to make them more accessible.

3. **PC 3 - Web Server (192.168.10.5/24 - RHEL)**
   Web Server provides web-based services, such as supermarket information systems.

4. **PC 4 - Database Server (192.168.10.6/24 - RHEL)**
   Database Server stores and manages important data such as stock of goods and transactions.

5. **PC 5 - Client Device (DHCP Client - Windows XP)**
   These devices are used by users to access services from existing servers.

CONFIGURING DHCP SERVER

1. Set the IP into static IP address for the server

```
[hengkel@localhost ~]$ nmcli connection modify ens160 ipv4.addresses 192.168.10.3/24
[hengkel@localhost ~]$ systemctl restart NetworkManager
[hengkel@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:97:8c:95 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.10.3/24 brd 192.168.10.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe97:8c95/64 scope link
       valid_lft forever preferred_lft forever
3: br0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 4a:eb:88:9f:e8:58 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global noprefixroute br0
       valid_lft forever preferred_lft forever
```

2. Download and install the DHCP

```
[hengkel@192 ~]$ sudo dnf install dhcp-server
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use subscription-manager to assign subscriptions.

Last metadata expiration check: 1:02:06 ago on Sat 22 Mar 2025 09:34:11 PM WIB.
Dependencies resolved.
================================================================================================================================================================
 Package                        Architecture              Version                                  Repository                                              Size
================================================================================================================================================================
Installing:
 dhcp-server                    x86_64                    12:4.4.2-19.b1.el9                       rhel-9-for-x86_64-baseos-rpms                          1.2 M

Transaction Summary
================================================================================================================================================================
Install  1 Package

Total download size: 1.2 M
Installed size: 3.9 M
Is this ok [y/N]: y
Downloading Packages:
dhcp-server-4.4.2-19.b1.el9.x86_64.rpm                                                                                   1.0 MB/s | 1.2 MB     00:01
----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                    1.0 MB/s | 1.2 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                                        1/1
  Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                                                                  1/1
  Installing       : dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                                                                  1/1
  Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                                                                  1/1
  Verifying        : dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                                                                  1/1
Installed products updated.

Installed:
  dhcp-server-12:4.4.2-19.b1.el9.x86_64

Complete!
```

CONFIGURING DHCP SERVER

3. Setting the DHCP Server

```
[hengkel@localhost ~]$ sudo nano /etc/dhcp/dhcpd.conf
```

```
option domain-name= "supermarketudin.com";
option domain-name-servers 192.168.10.4, 8.8.8.8;
default-lease-time 600;
max-lease-time 7200;

subnet 192.168.10.0 netmask 255.255.255.0 {
        range 192.168.10.50 192.168.10.70;
        option routers 192.168.10.1;
        option broadcast-address 192.168.10.255;
}
```

4. Check and see the IP in client PC

```
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:ea:02:f3 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.10.68/24 brd 192.168.10.255 scope global dynamic noprefixroute ens160
       valid_lft 567sec preferred_lft 567sec
    inet6 fe80::20c:29ff:feea:2f3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

CONFIGURING DNS SERVER

1. Set the IP into static IP address for the server

```
[hengkel@localhost ~]$ nmcli connection modify ens160 ipv4.addresses 192.168.10.4/24
[hengkel@localhost ~]$ systemctl restart NetworkManager
[hengkel@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:62:be:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.10.4/24 brd 192.168.10.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe62:be03/64 scope link
       valid_lft forever preferred_lft forever
3: br0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether ee:3e:56:0c:37:4c brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global noprefixroute br0
       valid_lft forever preferred_lft forever
[hengkel@localhost ~]$
```

2. Download and install the DNS

```
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                    1/1
  Installing       : bind-dnssec-doc-32:9.16.23-24.el9_5.3.noarch                                                       1/5
  Installing       : python3-ply-3.11-14.el9.noarch                                                                     2/5
  Installing       : python3-bind-32:9.16.23-24.el9_5.3.noarch                                                          3/5
  Installing       : bind-dnssec-utils-32:9.16.23-24.el9_5.3.x86_64                                                     4/5
  Running scriptlet: bind-32:9.16.23-24.el9_5.3.x86_64                                                                  5/5
  Installing       : bind-32:9.16.23-24.el9_5.3.x86_64                                                                  5/5
  Running scriptlet: bind-32:9.16.23-24.el9_5.3.x86_64                                                                  5/5
  Verifying        : python3-ply-3.11-14.el9.noarch                                                                     1/5
  Verifying        : bind-32:9.16.23-24.el9_5.3.x86_64                                                                  2/5
  Verifying        : bind-dnssec-doc-32:9.16.23-24.el9_5.3.noarch                                                       3/5
  Verifying        : bind-dnssec-utils-32:9.16.23-24.el9_5.3.x86_64                                                     4/5
  Verifying        : python3-bind-32:9.16.23-24.el9_5.3.noarch                                                          5/5
Installed products updated.

Installed:
  bind-32:9.16.23-24.el9_5.3.x86_64              bind-dnssec-doc-32:9.16.23-24.el9_5.3.noarch       bind-dnssec-utils-32:9.16.23-24.el9_5.3.x86_64
  python3-bind-32:9.16.23-24.el9_5.3.noarch      python3-ply-3.11-14.el9.noarch

Complete!
[hengkel@192 ~]$
```

3. Configurating named in `/etc/named.conf`

```
[hengkel@localhost ~]$ sudo nano /etc/named.conf
```

```
options {
        listen-on port 53 { 127.0.0.1;192.168.10.4; };
        listen-on-v6 port 53 { ::1; };
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        secroots-file   "/var/named/data/named.secroots";
        recursing-file  "/var/named/data/named.recursing";
        allow-query     { localhost; 192.168.10.4;any; };
        allow-query-cache {localhost; 192.168.10.4;any; };
```

CONFIGURING DNS SERVER

4. Configuring `/etc/named/supermarketudin.com.zone`

```
[hengkel@localhost ~]$ sudo nano /etc/named/supermarketudin.com.zone
```

```
$TTL    86400
@       IN      SOA     supermarketudin.com.    root.supermarketudin.com. (
                2018092501      ;Serial
                3600            ;Refresh
                1800            ;Retry
                604800          ;Expire
                86400           ;Minimum TTl
)

@       IN      NS      ns1.supermarketudin.com.
@       IN      NS      ns2.supermarketudin.com.
@       IN      A       192.168.10.4
ns1     IN      A       192.168.10.4
ns2     IN      A       192.168.10.4
www     IN      CNAME   supermarketudin.com.
blog    IN      A       192.168.10.4
```

5. Configuring `/etc/named/supermarketudin.com.rev`

```
[hengkel@localhost ~]$ sudo nano /etc/named/supermarketudin.com.rev
```

```
$TTL    86400
@       IN      SOA     supermarketudin.com.    root.supermarketudin.com. (
                2018092501      ;Serial
                3600            ;Refresh
                1800            ;Retry
                604800          ;Expire
                86400           ;Minimum TTl
)

        IN      NS      ns1.supermarketudin.com.
        IN      NS      ns2.supermarketudin.com.
@       IN      A       192.168.10.4
ns1     IN      A       192.168.10.4
ns2     IN      A       192.168.10.4
50      IN      PTR     supermarketudin.com.
50      IN      PTR     blog.supermarketudin.com.
```

CONFIGURING DNS SERVER

6. Restart, enble, and see the DNS status

```
[hengkel@localhost ~]$ systemctl restart named
[hengkel@localhost ~]$ systemctl enable named
[hengkel@localhost ~]$ systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
     Active: active (running) since Sat 2025-03-22 22:27:43 WIB; 13s ago
   Main PID: 4409 (named)
      Tasks: 26 (limit: 22795)
     Memory: 62.6M
        CPU: 84ms
     CGroup: /system.slice/named.service
             └─4409 /usr/sbin/named -u named -c /etc/named.conf

Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::b#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
Mar 22 22:27:43 localhost.localdomain named[4409]: network unreachable resolving './NS/IN': 2001:7fd::1#53
Mar 22 22:27:53 localhost.localdomain named[4409]: resolver priming query complete
Mar 22 22:27:53 localhost.localdomain named[4409]: managed-keys-zone: Unable to fetch DNSKEY set '.': timed out
```

7. Check if the DNS is working or not

```
[hengkel@localhost ~]$ dig supermarketudin.com

; <<>> DiG 9.16.23-RH <<>> supermarketudin.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 45372
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 31d0d23f7207cbbd0100000067ded72f242c9075c58c1c82 (good)
;; QUESTION SECTION:
;supermarketudin.com.           IN      A

;; ANSWER SECTION:
supermarketudin.com.    86400   IN      A       192.168.10.4

;; Query time: 2 msec
;; SERVER: 192.168.10.4#53(192.168.10.4)
;; WHEN: Sat Mar 22 22:28:47 WIB 2025
;; MSG SIZE  rcvd: 92
```

```
[hengkel@localhost ~]$ nslookup supermarketudin.com
Server:         192.168.10.4
Address:        192.168.10.4#53

Name:   supermarketudin.com
Address: 192.168.10.4
```

CONFIGURING WEB SERVER

1. Set the IP into static IP address for the server

```
[root@localhost abda]# nmcli connection modify enp0s3 ipv4.addresses 192.168.10.5/24
[root@localhost abda]# nmcli connection modify enp0s3 ipv4.gateway 192.168.10.1
[root@localhost abda]# nmcli connection modify enp0s3 ipv4.dns 8.8.8.8
[root@localhost abda]# nmcli connection modify enp0s3 ipv4.method manual
[root@localhost abda]# nmcli connection modify up enp0s3
Error: unknown connection 'up'.
[root@localhost abda]# nmcli connection  up enp0s3
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConn
ection/3)
```

2. Check if the IP has changed on demand

```
[root@localhost abda]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
 1000
    link/ether 08:00:27:b7:7c:51 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.5/24 brd 192.168.10.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb7:7c51/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

3. Starting Web Server (here we use http)

```
[root@localhost abda]# systemctl restart httpd
[root@localhost abda]# systemctl enable --now httpd
[root@localhost abda]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-03-23 09:37:40 WIB; 18s ago
       Docs: man:httpd.service(8)
   Main PID: 6740 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0
      Tasks: 177 (limit: 23017)
     Memory: 21.5M
        CPU: 101ms
     CGroup: /system.slice/httpd.service
             ├─6740 /usr/sbin/httpd -DFOREGROUND
             ├─6741 /usr/sbin/httpd -DFOREGROUND
             ├─6742 /usr/sbin/httpd -DFOREGROUND
             ├─6743 /usr/sbin/httpd -DFOREGROUND
             └─6744 /usr/sbin/httpd -DFOREGROUND

Mar 23 09:37:40 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Mar 23 09:37:40 localhost.localdomain httpd[6740]: AH00558: httpd: Could not reliably determine
Mar 23 09:37:40 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Mar 23 09:37:40 localhost.localdomain httpd[6740]: Server configured, listening on: port 80
```

CONFIGURING WEB SERVER

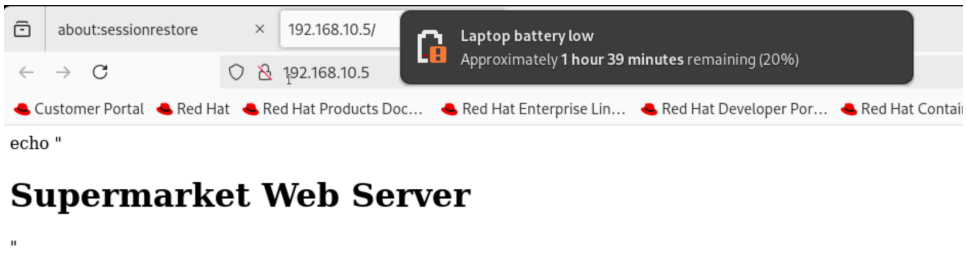
4. Add a service in the firewall

```
[root@localhost abda]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
[root@localhost abda]# firewall-cmd --reload
success
[root@localhost abda]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client dns http samba ssh
```

5. Creating a Simple Web Page as a Test of Whether It Works

```
[root@localhost html]# touch index.html
[root@localhost html]# nano index.html
[root@localhost html]# nano index.html
[root@localhost html]# cd
[root@localhost ~]# chown apache:apache /var/www/html/index.html
[root@localhost ~]# chmod 644 /var/www/html/index.html
[root@localhost ~]# curl http://192.168.10.5
echo "<h1>Supermarket Web Server</h1>"
```

6. Test in Browser



CONFIGURING DB SERVER

1. Installing Mariadb

```
[floraaulianilamaya@bender ~]$ sudo su
[sudo] password for floraaulianilamaya:
[root@bender floraaulianilamaya]# sudo dnf install -y mariadb-server
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can
use subscription-manager to assign subscriptions.

Red Hat Enterprise Linux 9 for x86_64 - BaseOS  5.5 kB/s | 4.1 kB     00:00
Red Hat Enterprise Linux 9 for x86_64 - AppStre 4.8 kB/s | 4.5 kB     00:00
Red Hat Enterprise Linux 9 for x86_64 - AppStre 1.2 MB/s |  52 MB     00:43
Package mariadb-server-3:10.5.27-1.el9_5.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Enabling and Running Mariadb

```
[root@bender floraaulianilamaya]# sudo systemctl enable --now mariadb
[root@bender floraaulianilamaya]# sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
 ... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] n
```

CONFIGURING DB SERVER

3. Securing Mariadb

```
 ... skipping.

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] Y
 ... Success!

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] Y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] Y
 - Dropping test database...
 ... Success!
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] Y
 ... Success!

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
```

4. Configuring a static IP using the nmcli command

```
[root@bender floraaulianilamaya]# nmcli device status
DEVICE  TYPE      STATE                   CONNECTION
ens160  ethernet  connected               ens160
lo      loopback  connected (externally)  lo
[root@bender floraaulianilamaya]# nmcli con mod ens160 ipv4.addresses 192.168.10.6/24
[root@bender floraaulianilamaya]# nmcli con mod ens160 ipv4.gateway 192.168.10.1
[root@bender floraaulianilamaya]# nmcli con mod ens160 ipv4.dns 8.8.8.8
[root@bender floraaulianilamaya]# nmcli con mod ens160 ipv4.method manual
[root@bender floraaulianilamaya]# nmcli con up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/Active
Connection/3)
```

CONFIGURING DB SERVER

5. Create Database and User

```
[root@bender floraaulianilamaya]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE supermarket_db;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> CREATE USER 'superuser'@'%' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON supermarket_db.* TO 'superuser'@'%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
```

6. Allow Remote Access

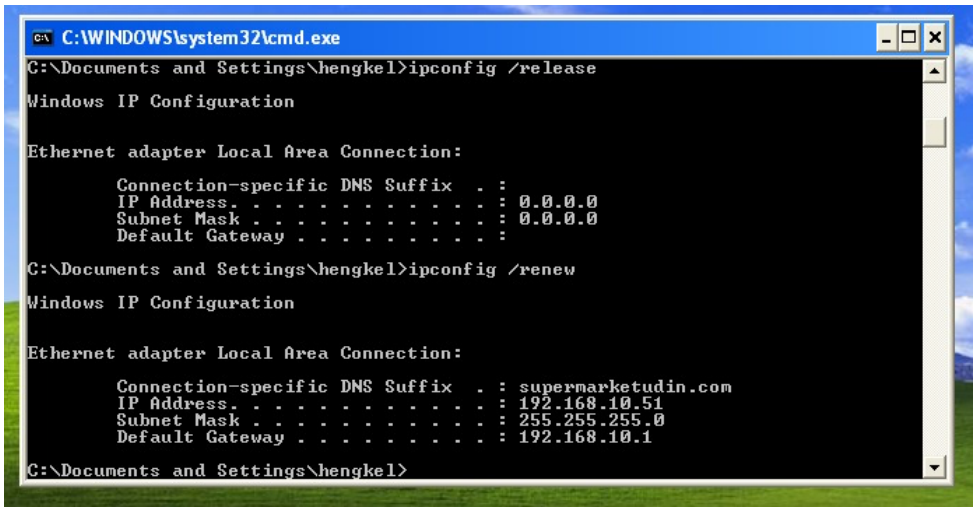
```
[root@bender floraaulianilamaya]# sudo sed -i 's/^bind-address.*/bind-address = 0.0.0.0/' /etc/my.cnf.d/mariadb-server.cnf
[root@bender floraaulianilamaya]# sudo systemctl restart mariadb
[root@bender floraaulianilamaya]# sudo firewall-cmd --add-service=mysql --permanent
success
[root@bender floraaulianilamaya]# sudo firewall-cmd --reload
success
```

7. Test Connection, Enter the password. If you successfully log in, the database server is ready.

```
[root@bender floraaulianilamaya]# mysql -h 192.168.10.6 -u superuser -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> EXIT;
Bye
[root@bender floraaulianilamaya]#
```

PC CLIENT TEST

Windows XP (as a client) Test



The results of connection tests on the client PC using `ipconfig /release` and `ipconfig /renew` indicate that the device successfully obtained an IP address dynamically from the DHCP server. Initially, the ipconfig /release command clears the IP configuration, while ipconfig /renew provides the IP address 192.168.10.51, the subnet mask 255.255.255.0, and the default gateway 192.168.10.1, indicating that the DHCP server is functioning properly. Additionally, the presence of a Connection-specific DNS Suffix with the domain `supermarketudin.com` suggests a network configuration connected to the internal domain. With this configuration, the client PC can communicate with other servers on the network, and further connectivity can be tested using ping commands to the default gateway or other servers.

REQUIREMENTS

Hardware :

1. ROG STRIX G16

Operating System:

1. Windows 11 64-bit
2. Redhat Enterprise Linux
3. Windows xp

Software :

1. Microsoft Word
2. VMWare
3. Oracle VirtualBox

PROJECT FILE DETAILS		
No	**Filename**	**Remarks**
1	project2group7.docx	Microsoft Words contain research paper about the project
2	project2group7.pdf	Paper file